

SFLMAC: Mitigating Stragglers in Split Federated Learning with Main-Aux Cooperation

Wenwu Hou¹, Wenxuan Zhou¹, Ninghui Jia¹, and Zhihao Qu¹

Hohai University, Nanjing, China
{wwhou,zhouwx,hhu_jnh,qzhihao}@hhu.edu.cn

Abstract. The straggler problem is a significant challenge in split federated learning (SFL) due to the high heterogeneity of client resources. Previous approaches address this by adopting asynchronous updates to eliminate client synchronization delays. However, this introduces the staleness problem, which can impair model convergence due to outdated updates from slower clients. To address this, we propose SFLMAC, a split federated collaborative learning framework that strategically pairs resource-rich main clients with weaker auxiliary clients. In this asymmetric design, main clients retain the full model while auxiliary clients only host their respective bottom models. Crucially, auxiliary clients update their local bottom models and simultaneously contribute gradients to their corresponding main client’s top model, while main clients train their full models on local data. This heterogeneity-aware pairing and workload partitioning drastically reduces the computational burden on weak clients, accelerates global synchronization, and effectively mitigates the straggler problem. The server aggregates models layer by layer, reducing communication overhead by roughly half compared with standard SFL. We provide a convergence analysis and show that SFLMAC achieves a convergence rate comparable to standard SFL. Extensive experiments on various models and datasets demonstrate that SFLMAC outperforms state-of-the-art methods. The full appendix can be found in <https://github.com/Hww76/WASA2026.git>.

Keywords: Split Federated Learning · Client Heterogeneity · Straggler.

1 Introduction

With the rapid adoption of smart homes and Internet-of-Things (IoT) applications, edge devices continuously generate and store large amounts of privacy-sensitive data [21]. These data are valuable for intelligent services as they remain distributed across local devices and are difficult to centralize [2]. To exploit such data without violating privacy constraints, split federated learning (SFL) has emerged as a practical paradigm for collaborative model training at edge environment [18].

SFL reduces the burden of full local training by partitioning the model across clients and the server [9, 10]. In a typical SFL system, clients optimize the bottom model on local data, while the server maintains the top model. To complete forward and backward propagation, clients need to continuously exchange smashed

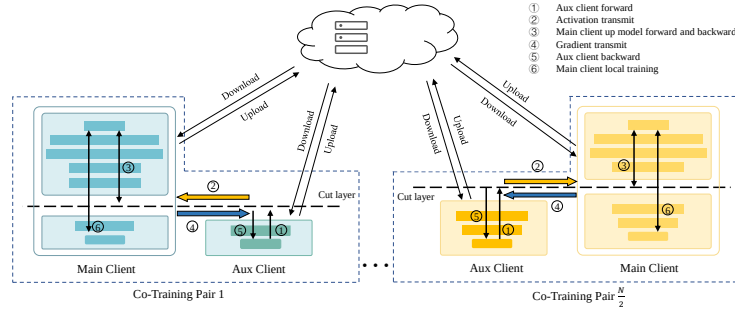


Fig. 1: Illustration of our proposed SFLMAC.

data and gradients with the server [16, 22]. However, participating devices in real deployments often differ greatly in computation capability [5, 17], which leads to stragglers that slow down synchronous training [6, 13]. Traditional asynchronous update methods have been explored to mitigate this issue, but they suffer from staleness, where outdated updates from slower clients may hinder or even prevent model convergence [3].

To mitigate the straggler problem of SFL, recent studies have explored client-collaborative split training [14, 19, 24]. Despite these efforts, existing client-collaborative SFL approaches still face key limitations, including restricted per-round data coverage, new communication bottlenecks, and increased coordination complexity. To address these limitations, we propose SFLMAC (split federated learning with main-aux cooperation), a client-collaborative SFL framework for computing-heterogeneous edge environments. Specifically, SFLMAC groups clients into main-auxiliary pairs according to their resource levels. In each pair, the resource-rich main client maintains the full model, while the weaker auxiliary client only maintains the bottom model. The main and auxiliary clients independently update their own bottom models, while collaboratively update the top model of the main client. Compared with standard SFL, the smashed data generated by the main client can be consumed locally, thereby significantly reducing communication overhead. By offloading most computation from weak clients to strong ones, SFLMAC reduces the burden on resource-constrained devices, alleviates stragglers, and reduces global synchronization delay.

The main contributions of this work are summarized as follows:

- We propose SFLMAC, a client-collaborative SFL framework that organizes clients into main-auxiliary pairs. This design alleviates the straggler problem in heterogeneous edge environments while significantly reducing communication overhead by roughly half compared with standard SFL.
- We provide a theoretical convergence analysis for SFLMAC, showing that the proposed alternating update scheme does not slow down convergence and that SFLMAC attains a convergence rate comparable to standard SFL.
- We conduct extensive experiments to verify the effectiveness of SFLMAC, showing that it achieves a favorable trade-off among training efficiency, communication cost, and model performance compared with representative baselines.

2 Related Work

Split federated learning (SFL) splits neural networks between clients and a server to enable large-scale model training on resource-constrained edge devices [22]. However, heterogeneous edge environments and frequent exchange of smashed data make SFL vulnerable to the straggler effect and communication bottleneck.

Several recent studies have attempted to mitigate the straggler problem in SFL. MU-SplitFed [13] reduces server idle time through unbalanced server-client updates, Fed-SASL [20] and SAFSL [1] introduces asynchronous computation to relax strict synchronization, and heterogeneity-aware collaborative schemes such as CSFL [6] and HASFL [7] alleviate synchronization delay by redistributing computation or adaptively adjusting training workloads.

Some studies have attempted to alleviate the server-side communication bottleneck in SFL through collaborative training among clients. PairingFL [24] divides clients into a bottom model group and an upper model group, where each training round is carried out using the local data of clients in the bottom model group. ParallelSFL [14] groups clients into clusters, where one client in each cluster maintains the top model and the remaining clients maintain the bottom models, and these bottom and top models are collaboratively trained within the cluster. RingSFL [19] organizes clients in a ring topology, where the computation is distributed across clients according to their resource capacities and collaboratively executed for each sample.

Despite these efforts, existing SFL approaches still suffer from several common limitations. Methods designed to mitigate stragglers often reduce training time at the cost of model accuracy. Existing client-collaborative methods still suffer from single-point bottlenecks, inefficient training, and strong dependency among clients. As a result, existing solutions have not yet achieved a satisfactory balance among straggler mitigation, communication efficiency, and training efficiency.

3 Preliminary

3.1 Split Federated Learning

We consider an SFL system with N clients, where client i owns a local dataset D_i . To reduce the burden of full local training, the model is split at a cut layer into a client-side submodel and a server-side submodel. For an input sample ξ , the prediction model is written as

$$f(\mathbf{w}; \xi) = f_s(\mathbf{w}_s; f_c(\mathbf{w}_c; \xi)), \quad (1)$$

where $\mathbf{w} = (\mathbf{w}_c, \mathbf{w}_s)$, with $\mathbf{w}_c$ and $\mathbf{w}_s$ denoting the parameters of the client-side and server-side submodels, respectively.

The global training objective is

$$\min_{\mathbf{w}_c, \mathbf{w}_s} F(\mathbf{w}_c, \mathbf{w}_s) := \sum_{i=1}^N \frac{|D_i|}{D} F_i(\mathbf{w}_c, \mathbf{w}_s), \quad (2)$$

where $D = \sum_{i=1}^N |D_i|$, and

$$F_i(\mathbf{w}_c, \mathbf{w}_s) = \mathbb{E}_{(\xi, y) \sim D_i} [\ell(f_s(\mathbf{w}_s; f_c(\mathbf{w}_c; \xi)), y)]. \quad (3)$$

In communication round t , client i first computes the client-side representation

$$\mathbf{a}_i^t = f_c(\mathbf{w}_{c,i}^t; \xi_i), \quad (4)$$

where $\mathbf{a}_i^t$ is the smashed data at the cut layer. The smashed data are transmitted to the main server, which performs forward propagation on the server-side submodel, computes the loss, and obtains the gradient with respect to the smashed data,

$$\mathbf{g}_{a,i}^t = \nabla_{\mathbf{a}_i^t} \ell(f_s(\mathbf{w}_s^t; \mathbf{a}_i^t), y_i). \quad (5)$$

The gradient $\mathbf{g}_{a,i}^t$ is then sent back to client i to complete client-side backpropagation and update the local client-side submodel. After local training, the fed server aggregates the client-side submodels as

$$\mathbf{w}_c^{t+1} = \sum_{i=1}^N \frac{|D_i|}{D} \mathbf{w}_{c,i}^{t+1}, \quad (6)$$

while the server-side submodel is maintained and updated on the main server.

4 Method

In this section, we present the client pairing scheme, adaptive cut layer selection, and training procedure of SFLMAC. Clients are sorted by computational capacity and divided into a main group and an auxiliary group, each of size P . The overall procedure is summarized in Algorithm 1.

4.1 Main-Auxiliary Pair Formation

Client Grouping by Computational Capacity Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of clients. For each client $i \in \mathcal{N}$, let $c_i > 0$ denote its computational capacity, where a larger c_i indicates a stronger client. The clients are sorted in ascending order of computational capacity, i.e.,

$$c_{(1)} \leq c_{(2)} \leq \dots \leq c_{(N)}, \quad (7)$$

where $c_{(k)}$ denotes the computational capacity of the client ranked at position k . Based on the sorted order, all clients are divided into two groups: the main group $\mathcal{G}_m$ and the auxiliary group $\mathcal{G}_a$. Specifically, the stronger half of the clients are assigned to $\mathcal{G}_m$, while the weaker half are assigned to $\mathcal{G}_a$.

Algorithm 1 SFLMAC Training Procedure

Require: Initial global parameters $\mathbf{w}_1$, learning rate η , local step number K , communication round number T

Ensure: Final global parameters $\mathbf{w}_T$,

- 1: Sort all clients by computational capacity and divide them into the main group and auxiliary group, each of size P
- 2: Construct the paired client set $\mathcal{P} = \{(m_i, a_i)\}_{i=1}^P$ according to **Pairing Strategy**
- 3: **for** each pair $i = 1, 2, \dots, P$ **do**
- 4: Select the cut layer l_i^* for pair i according to Eq. (10)
- 5: **end for**
- 6: **for** $t = 1, 2, \dots, T$ **do**
- 7: **for** each pair $i = 1, 2, \dots, P$ **in parallel do**
- 8: Receive the split global parameters $(\mathbf{v}_{i,t}, \mathbf{u}_{i,t})$ from the server
- 9: Initialize local parameters $\mathbf{v}_{i,t}^{m,1} = \mathbf{v}_{i,t}$, $\mathbf{v}_{i,t}^{a,1} = \mathbf{v}_{i,t}$, and $\mathbf{u}_{i,t}^1 = \mathbf{u}_{i,t}$
- 10: **for** $k = 1, 2, \dots, K$ **do**
- 11: Auxiliary client updates using Eq. (11)
- 12: Main client updates using Eq. (12)
- 13: **end for**
- 14: Upload $\mathbf{u}_{i,t}^K$, $\mathbf{v}_{i,t}^{m,K}$, and $\mathbf{v}_{i,t}^{a,K}$ to the server
- 15: **end for**
- 16: Server updates the global model using Eq. (14)
- 17: **end for**

Optimization Objective Consider a model with L layers and let $l \in \{1, \dots, L-1\}$ denote the cut layer. For each layer l , let ϕ_l^f and ϕ_l^b denote the computation cost of forward propagation and backward propagation, respectively. The total computation cost of layer l is defined as

$$\phi_l = \phi_l^f + \phi_l^b. \quad (8)$$

For a main-auxiliary pair, let c_{main} and c_{aux} denote the computational capacities of the main and auxiliary clients, respectively, with $c_{\text{main}} > c_{\text{aux}}$. The auxiliary client only updates the bottom model, whereas the main client trains the full model on its own data and additionally computes the top model for the auxiliary client. To balance the workloads between the main and auxiliary clients, we first quantify their respective workloads under cut layer l , and then select the cut layer by minimizing the bottleneck load within the pair.

$$l^* = \arg \min_{1 \leq l \leq L-1} \max \left\{ \sum_{k=1}^l \frac{\phi_k}{c_{\text{aux}}}, \sum_{k=1}^L \frac{\phi_k}{c_{\text{main}}} + \sum_{k=l+1}^L \frac{\phi_k}{c_{\text{main}}} \right\}. \quad (9)$$

Pairing Strategy Since the straggler effect is determined by the weakest clients, the pairing strategy should prioritize reducing their burden. Only the strongest main client can take over more top-model computation without becoming a new bottleneck.

This motivates a strongest-to-weakest pairing rule: the weakest auxiliary client is paired with the strongest main client, and the remaining clients are matched in the same manner until all pairs are formed.

This pairing rule is optimal for the considered objective, and detailed theoretical derivations and proofs are provided in Appendix C of the online supplementary material [8]. While our base pairing strategy assumes static client capabilities, we have designed it to handle real-world edge dynamics and edge cases. In practice, client computing power may fluctuate, network connections may be interrupted, and clients may drop out unexpectedly, which can invalidate the initial pairing order or prevent auxiliary clients from completing training. To address these issues, we perform a global re-pairing at the start of each communication round, based on the latest client online status and computational capability measurements. For an odd number of clients, the strongest remaining client operates as a standalone full-model trainer.

Pair-wise Adaptive Cut Layer Selection For each main-auxiliary pair, the cut layer is selected by traversing the candidate cut layer from deep to shallow layers. l^* is initialized with $L - 1$, the resulting cut layer is taken as the optimal one, i.e.,

$$\text{while } (l^* \geq 1 \text{ and } W_{\text{aux}}(l^*) > W_{\text{main}}(l^*)), \quad l^* \leftarrow l^* - 1. \quad (10)$$

Thus, l^* is the deepest feasible cut layer such that the auxiliary client does not become the bottleneck.

4.2 One Local Synchronization Step

For pair i in global iteration t and local step k , let $\mathbf{u}_{i,t}^k$ denote the shared top model of pair i , and let $\mathbf{v}_{i,t}^{a,k}$ and $\mathbf{v}_{i,t}^{m,k}$ denote the bottom model of the auxiliary and main clients, respectively. Let η be the local learning rate. Moreover, let $\mathbf{g}_{i,t}^{a,k}$ and $\mathbf{g}_{i,t}^{m,k+\frac{1}{2}}$ denote the stochastic gradients of the auxiliary and main clients with respect to the shared up model, and let $\mathbf{h}_{i,t}^{a,k}$ and $\mathbf{h}_{i,t}^{m,k}$ denote their stochastic gradients with respect to the corresponding bottom model.

Paired Cooperation Update SFLMAC organizes local training around fixed main-auxiliary pairs. Within each pair, the two clients maintain their own lower-layer parameters while collaboratively optimizing a shared upper-layer.

At the k -th local step of pair i in the t -th global iteration, the auxiliary client first updates the shared top model and its own bottom model as

$$\mathbf{u}_{i,t}^{k+\frac{1}{2}} = \mathbf{u}_{i,t}^k - \eta \mathbf{g}_{i,t}^{a,k}, \quad \mathbf{v}_{i,t}^{a,k+1} = \mathbf{v}_{i,t}^{a,k} - \eta \mathbf{h}_{i,t}^{a,k}, \quad (11)$$

where $\mathbf{g}_{i,t}^{a,k}$ and $\mathbf{h}_{i,t}^{a,k}$ denote the top model and bottom model stochastic gradients of the auxiliary client, respectively. The main client then continues from the intermediate upper-layer state $\mathbf{u}_{i,t}^{k+\frac{1}{2}}$ and performs

$$\mathbf{u}_{i,t}^{k+1} = \mathbf{u}_{i,t}^{k+\frac{1}{2}} - \eta \mathbf{g}_{i,t}^{m,k+\frac{1}{2}}, \quad \mathbf{v}_{i,t}^{m,k+1} = \mathbf{v}_{i,t}^{m,k} - \eta \mathbf{h}_{i,t}^{m,k}, \quad (12)$$

where $\mathbf{g}_{i,t}^{m,k+\frac{1}{2}}$ and $\mathbf{h}_{i,t}^{m,k}$ are the corresponding stochastic gradients of the main client.

Combining the two updates, the shared upper-layer evolves as

$$\mathbf{u}_{i,t}^{k+1} = \mathbf{u}_{i,t}^k - \eta \mathbf{g}_{i,t}^{a,k} - \eta \mathbf{g}_{i,t}^{m,k+\frac{1}{2}}. \quad (13)$$

Therefore, one local synchronization step in SFLMAC consists of two consecutive updates contributed by the auxiliary and main clients, respectively, while the lower-layer parameters remain client-specific throughout local training.

Global Aggregation After K local iterations, the server aggregates the updated parameters from all pairs to obtain the next global model:

$$\mathbf{w}_{t+1} = \sum_{i=1}^P \frac{1}{P} \left(\frac{1}{2} \mathbf{v}_{i,t}^{a,K} + \frac{1}{2} \mathbf{v}_{i,t}^{m,K} + \mathbf{u}_{i,t}^K \right) \quad (14)$$

4.3 Training Procedure

In each global iteration, the server broadcasts to each pair the corresponding lower-layer and upper-layer parameters, denoted by $(\mathbf{v}_{i,t}, \mathbf{u}_{i,t})$ according to its cut layer. Then, each main-auxiliary pair performs K local collaborative steps, during which the main and auxiliary clients alternately update the shared upper-layer while maintaining their own lower-layer parameters, as illustrated in Fig. 1. After local training, the server collects the resulting bottom and top models from all pairs and aggregates them to obtain the next global model. Thus, the server is only responsible for global coordination and aggregation, which alleviates the server-side communication bottleneck by removing its involvement in step-wise upper-layer optimization within each client.

5 Convergence Analysis

In this section, we provide the convergence analysis and theoretical insights for the proposed framework. Specifically, we introduce the assumptions on the loss function and the Non-Convex convergence analysis of the proposed framework.

We make the following standard assumptions, which are widely adopted in distributed optimization [11, 23].

Assumption 1 (Lipschitz Smoothness) *There exists a constant $L > 0$ such that, for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$:*

$$\|\nabla F(\mathbf{x}_1) - \nabla F(\mathbf{x}_2)\| \leq L \|\mathbf{x}_1 - \mathbf{x}_2\|. \quad (15)$$

Assumption 2 (Bounded Gradients) *There exists $G_u > 0$ is constant, such that for any model parameter $\mathbf{x} = (\mathbf{v}, \mathbf{u}) \in \mathbb{R}^d$, the block gradients are bounded as:*

$$\|\nabla_{\mathbf{u}} F(\mathbf{x})\| \leq G_u. \quad (16)$$

Assumption 3 (Unbiased Gradients with Bounded Variance) *There exists $\sigma_v > 0$ and $\sigma_u > 0$ are constants, such that for any device i , global round t , and local iteration k , the stochastic gradients with respect to the bottom and top model, computed from a random sample $\xi_{i,t}^k \sim D_i$, are unbiased estimators of the corresponding global gradients at $\mathbf{x}_t$:*

$$\mathbb{E}_{\xi_{i,t}^k \sim D_i} [h_i(\mathbf{x}_t; \xi_{i,t}^k)] = \nabla_v F(\mathbf{x}_t), \quad \mathbb{E}_{\xi_{i,t}^k \sim D_i} [g_i(\mathbf{x}_t; \xi_{i,t}^k)] = \nabla_u F(\mathbf{x}_t). \quad (17)$$

Moreover, the variances of the stochastic gradients are bounded as

$$\mathbb{E}_{\xi_{i,t}^k \sim D_i} [\|h_i(\mathbf{x}_t; \xi_{i,t}^k) - \nabla_v F(\mathbf{x}_t)\|^2] \leq \sigma_v^2, \quad \mathbb{E}_{\xi_{i,t}^k \sim D_i} [\|g_i(\mathbf{x}_t; \xi_{i,t}^k) - \nabla_u F(\mathbf{x}_t)\|^2] \leq \sigma_u^2 \quad (18)$$

Theorem 1. *Under the assumptions 1-3, when the step size $\eta \leq \frac{1}{4LK}$, and $\eta = \sqrt{\frac{2(F(\mathbf{x}_0) - F(\mathbf{x}_T))N}{(\sigma_v^2 + 3\sigma_u^2)LKT}}$, where K is local iterations. SFLMAC enjoys the following convergence rate:*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 \leq \frac{2(F(\mathbf{x}_0) - F(\mathbf{x}_T))}{\eta KT} + 2C_t. \quad (19)$$

where

$$C_t = \frac{L}{2N} \eta (\sigma_v^2 + 3\sigma_u^2) + \frac{L^2}{2} \eta^2 (\sigma_u^2 + G_u^2) + \frac{L^3}{2} \eta^3 (\sigma_u^2 + G_u^2) \left(K + \frac{3}{N}\right). \quad (20)$$

We observe that the η^2 and η^3 terms, which include the component explicitly introduced by SFLMAC's collaborative alternating update, are higher-order terms. It decays faster than the leading $O(T^{-1/2})$ rate as the number of communication rounds T increases.

For sufficiently large T , these higher-order terms become asymptotically negligible, so the upper bound in Eq. (19) is dominated by the leading term, yielding $\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 \preceq O(1/\sqrt{T})$.

Thus, SFLMAC preserves the standard $O(1/\sqrt{T})$ convergence rate of non-convex stochastic first-order methods, with the collaborative update mechanism only contributing negligible higher-order terms that do not impact the leading-order convergence performance. Detailed theoretical derivations and proofs are provided in Appendix B of the online supplementary material [8].

6 Experiments

6.1 Experimental Settings

Experimental Setup Unless otherwise specified, the number of clients is set to 20, and all clients participate in each communication round. For a fair comparison, all methods perform local training on one minibatch in each local iteration. Each client runs 64 local iterations with a learning rate of 0.01 and a minibatch size of 64. We adopt AlexNet, VGG16, and ResNet18 as the backbone models. To simulate heterogeneous client resources, the computational capability of each client is randomly assigned within the range from 10^7 to 10^9 FLOPs.

Table 1: Test Accuracy (%) of Different Methods under Different Numbers of Clients on CIFAR-10 after 50 Communication Rounds.

Method	AlexNet			VGG16			ResNet18		
	$N = 8$	$N = 16$	$N = 32$	$N = 8$	$N = 16$	$N = 32$	$N = 8$	$N = 16$	$N = 32$
FedAvg [15]	69.99	70.53	70.91	84.51	84.75	85.20	83.94	84.91	85.10
SplitFed [22]	<u>78.04</u>	<u>79.32</u>	<u>80.28</u>	<u>87.00</u>	86.68	86.58	88.13	<u>88.02</u>	86.85
RingSFL [19]	69.43	70.70	71.27	85.93	<u>86.85</u>	<u>86.76</u>	<u>86.63</u>	87.14	86.80
SFLMAC (Ours)	78.97	80.31	80.49	87.84	88.88	89.08	88.67	89.53	89.79

Baselines We compare our method with, FedAvg [15], SplitFed [22], and RingSFL [19]. FedAvg is a classical federated learning method. SplitFed is a server-side model that is centrally. RingSFL is a split federated learning method designed for heterogeneous clients and client-side model central settings.

Data Partitioning We conduct our main experiments on CIFAR-10 [12]. To further validate the effectiveness of SFLMAC, we additionally perform extensive experiments under various settings on CIFAR-100 [12], and CINIC-10 [4]. We adopt a label distribution partition based on the Dirichlet distribution. Specifically, we set the Dirichlet parameter α to 5 and 0.2, which correspond to approximately IID and clearly Non-IID settings, respectively.

6.2 Performance Evaluation

We first evaluate the convergence performance of SFLMAC and analyze the theoretical communication cost. Subsequently, we evaluate the performance in straggler mitigation and the impact of the shared top model.

Convergence Performance Table 1 summarizes the test accuracy of five methods on CIF-10 after 50 communication rounds, using AlexNet, VGG16, and ResNet18 under different client numbers. In general, the test accuracy of all methods improves with the increase in client quantity. The proposed SFLMAC outperforms FedAvg, SplitFed, and RingSFL by a clear margin across all setups.

As illustrated in Fig. 2, Fig. 3 and Fig. 4, SFLMAC consistently achieves the highest final Top-1 accuracy across different network architectures and datasets. Notably, while SplitFed exhibits the fastest initial convergence, it struggles to achieve the best final accuracy, with performance plateauing or degrading significantly under severe Non-IID settings (e.g., CIFAR-100 and CINIC-10).

This fast convergence of SplitFed stems from its centralized upper model, which learns from the concatenated activations of all clients simultaneously. However, this centralized mechanism is also its bottleneck: aggregating highly heterogeneous activations into a single server-side mini-batch causes severe gradient conflicts and representation interference. In contrast, SFLMAC optimizes a pair-wise joint objective locally. By constraining the main and auxiliary clients to alternately update a shared upper model in a sequential relay manner, SFLMAC effectively smooths out the local data heterogeneity and aligns feature representations within

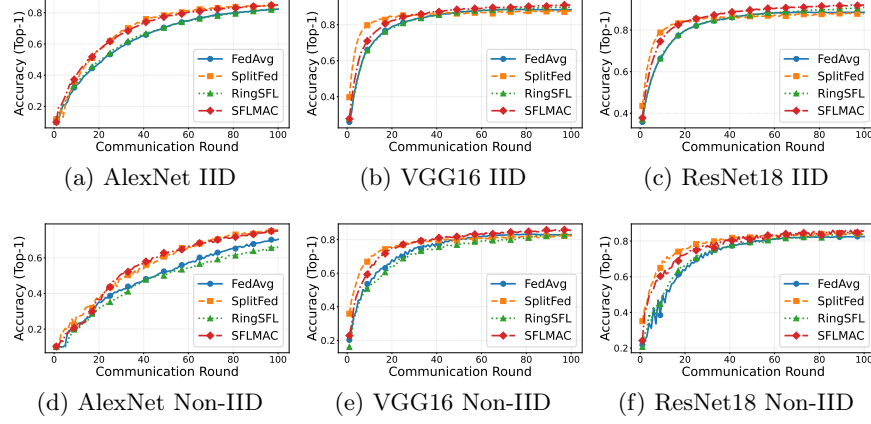


Fig. 2: Testing accuracy of different models on CIFAR-10(IID/Non-IID) under different algorithms.

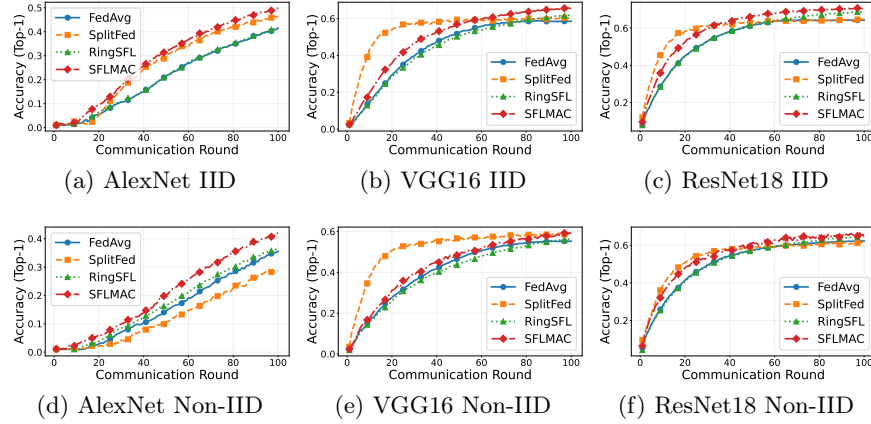


Fig. 3: Testing accuracy of different models on CIFAR-100(IID/Non-IID) under different algorithms.

each pair before global aggregation, thereby yielding superior robustness and higher final accuracy.

Specifically, SFLMAC’s superior performance is attributed to its tailored design. Its main-auxiliary client pairing enables localized alternating updates on the shared top model, thereby strengthening the consistency of global feature extraction and boosting model generalization.

Overall, SFLMAC achieves a favorable trade-off between adaptation to heterogeneous systems and model accuracy, making it more practical for real-world split federated learning deployments.

Communication Cost Analysis Table 2 provides a theoretical analysis of the communication overhead per global round for all methods, quantified by

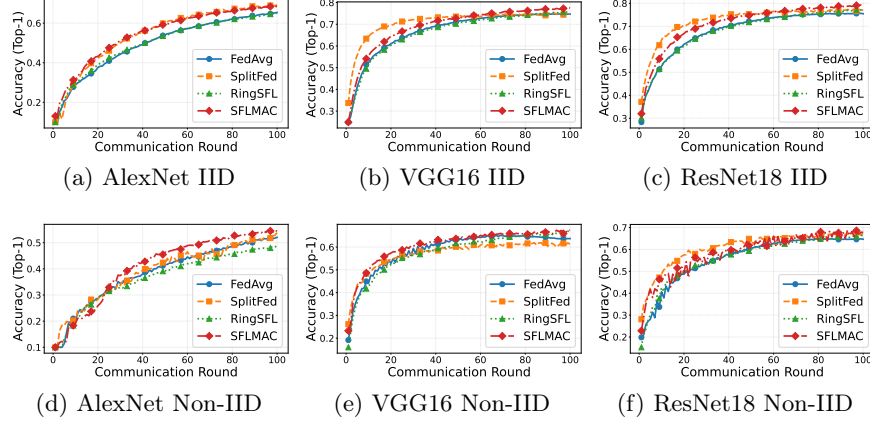


Fig. 4: Testing accuracy of different models on CINIC-10(IID/Non-IID) under different algorithms.

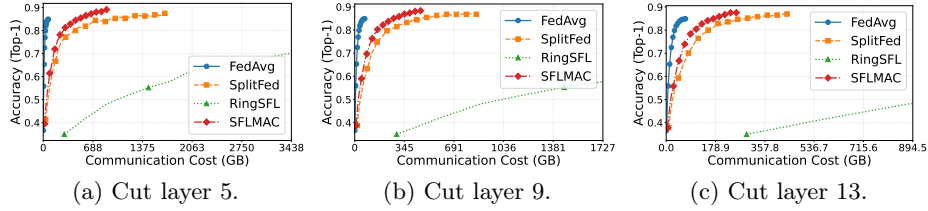


Fig. 5: Impact of cut layer on different algorithms with ResNet18 on CIFAR-10 after 50 Communication Rounds.

both the average cost per-client and the total cost of the entire system. In this table, N denotes the total number of clients, K represents the number of local iterations, $\bar{Q}$ is the average size of intermediate activations per minibatch, and $|w| = |w_l| + |w_u|$, where $|w_l|$ and $|w_u|$ correspond to the sizes of the lower-layer and upper-layer submodel respectively. Unlike SplitFed where all clients send activations to the server, SFLMAC only has auxiliary clients send activations to their paired main clients. Main clients compute locally, cutting activation transmission by half and total cost by 50%.

The proposed SFLMAC achieves a better accuracy-communication trade-off. In Fig. 5, FedAvg shows the lowest communication cost, but it requires full-model

Table 2: Communication Cost Analysis

Method	Communication Cost (Per Global Round)	
	Per Client (Average)	Total System
FedAvg [15]	$2 \mathbf{w} $	$2N \mathbf{w} $
SplitFed [22]	$2K\bar{Q} + 2 \mathbf{w}_l $	$2N(K\bar{Q} + \mathbf{w}_l)$
RingSFL [19]	$2NK\bar{Q} + 2 \mathbf{w} $	$2N^2K\bar{Q} + 2N \mathbf{w} $
SFLMAC (Ours)	$K\bar{Q} + 2 \mathbf{w}_l + \mathbf{w}_u $	$N(K\bar{Q} + 2 \mathbf{w}_l + \mathbf{w}_u)$

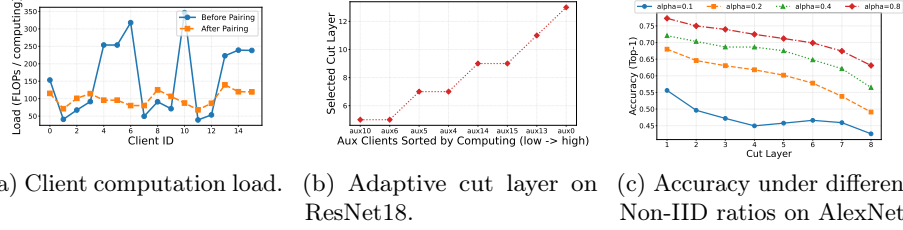


Fig. 6: Cut Layer Impact on SFLMAC: Load Balancing, Heterogeneity-Adaptive Split, and Non-IID Robustness.

training on clients and is impractical for real-world scenarios. SFLMAC consumes roughly half the communication overhead of SplitFed while achieving comparable accuracy. RingSFL requires excessive communication overhead due to its training paradigm.

Straggler Mitigation Fig. 6 (a) demonstrates the effectiveness of the proposed pairing strategy. Through pairwise collaborative training, the computational load across clients becomes substantially more balanced. Nevertheless, due to the fact that the cut layer is selected from a discrete set of model layers, perfectly equalized workloads cannot be achieved.

Fig. 6 (b) validates the rationality of the pairing design. To fully offload the training burden of weak-computing clients, it is essential to pair them with the most powerful computing devices for collaborative training. As shown in the figure, as the computing capacities of main-auxiliary client pairs become increasingly balanced, the internal cut layer moves upward continuously.

Ablation Study on Cut Layers Fig. 6 (c) shows the results with a fixed cut layer for all client pairs. In this figure, α represents the degree of data Non-IID, where a smaller α indicates a more severe Non-IID distribution. When the cut layer is equal to 8, it means there is no model splitting, which degenerates the framework to the traditional FedAvg training mode. As clearly observed, model accuracy consistently rises as the cut layer becomes shallower. This result directly validates our core design: both main and auxiliary clients train on their own local data, and each contributes their unique data distribution to the shared top model. The shared top model learns from the combined data of both clients in each pair, achieving stronger generalization. This demonstrates that more shared upper layers bring stronger model generalization performance. This verifies that the strategy of sharing the top model not only effectively mitigates the straggler problem but also delivers additional training benefits.

7 Conclusion

In this paper, we have proposed SFLMAC, a framework designed to overcome the straggler issues and communication bottlenecks in heterogeneous edge envi-

ronments. Rather than relying on a central server for top model computations, SFLMAC groups clients into main-auxiliary pairs to perform localized, alternating updates on a shared top model, effectively balancing computational load across heterogeneous edge devices and achieving more training efficiency. Empirical results confirm that SFLMAC achieves competitive accuracy and faster convergence while significantly reducing communication overhead compared to previous methods.

For future work, we will extend the framework to handle extreme heterogeneity scenarios where most devices are resource-constrained, by allowing main clients to support multiple auxiliary clients. We will also further optimize the framework for highly dynamic edge environments with changing network conditions.

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A Notation

For each shared upper-layer model, we equivalently regard the corresponding main and auxiliary clients as holding identical top model in the theoretical analysis. Let the global model at communication round t be denoted by $\mathbf{x}_t = (\mathbf{v}_t, \mathbf{u}_t)$, where $\mathbf{v}_t$ and $\mathbf{u}_t$ represent the bottom and top model, respectively. For device i , let $F_i(\mathbf{x})$ denote its local objective function with $\mathbf{x} = (\mathbf{v}, \mathbf{u})$, and define the global objective function as

$$F(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N (F_i(\mathbf{x}_i)). \quad (21)$$

At global round t , device i performs K local iterations indexed by $k = 1, \dots, K$.

We denote the local model at the k -th local iteration by $\mathbf{x}_{i,t}^k = (\mathbf{v}_{i,t}^k, \mathbf{u}_{i,t}^k)$, and let $\boldsymbol{\xi}_{i,t}^k$ denote the random sample drawn at this local state. The stochastic gradients of device i with respect to $\mathbf{v}$ and $\mathbf{u}$ are denoted by $\mathbf{h}_i(\mathbf{x}_{i,t}^k; \boldsymbol{\xi}_{i,t}^k)$ and $\mathbf{g}_i(\mathbf{x}_{i,t}^k; \boldsymbol{\xi}_{i,t}^k)$, respectively.

The corresponding true gradients are denoted by $\nabla_{\mathbf{v}} F_i(\mathbf{x}_{i,t}^k)$ and $\nabla_{\mathbf{u}} F_i(\mathbf{x}_{i,t}^k)$ for the local objective, and by $\nabla_{\mathbf{v}} F(\mathbf{x}_t)$ and $\nabla_{\mathbf{u}} F(\mathbf{x}_t)$ for the global objective.

B Convergence Analysis

B.1 Pair-level cumulative update.

The N devices are partitioned into P collaborative pairs, where $N = 2P$. For the p -th pair, denote the main-side and auxiliary-side bottom model by $\mathbf{v}_{p,t}^m$ and $\mathbf{v}_{p,t}^a$, respectively, and the shared upper-layer parameter by $\mathbf{u}_{p,t}$. We define the pair-level cumulative update at round t as

$$Q_{p,t} := (Q_{p,t}^{m,v}, Q_{p,t}^{a,v}, Q_{p,t}^u), \quad (22)$$

where

$$Q_{p,t}^{m,v} := \sum_{k=1}^K h_{p,t}^{m,k}, \quad Q_{p,t}^{a,v} := \sum_{k=1}^K h_{p,t}^{a,k}, \quad (23)$$

and

$$Q_{p,t}^u := \sum_{k=1}^K (g_{p,t}^{a,k} + g_{p,t}^{m,k} - r_{p,t}^k). \quad (24)$$

where

$$r_{p,t}^k := g_{p,t}^{m,k} - g_{p,t}^{m,k+\frac{1}{2}} \quad (25)$$

Accordingly, the global update can be written as

$$\mathbf{x}_{t+1} - \mathbf{x}_t = -\frac{\eta}{N} \sum_{p=1}^P Q_{p,t}. \quad (26)$$

For convenience, define the conditional mean

$$\bar{Q}_{p,t} := \mathbb{E}[Q_{p,t} \mid \mathbf{x}_t]. \quad (27)$$

B.2 Non-Convex Convergence Analysis

Lemma 1 (Pair-level mean decomposition). *Under Assumption 3, for each collaborative pair p and round t , the conditional mean $\bar{Q}_{p,t}$ satisfies*

$$\bar{Q}_{p,t} = \left(K \nabla_{\mathbf{v}} F(\mathbf{x}_t), K \nabla_{\mathbf{v}} F(\mathbf{x}_t), 2K \nabla_{\mathbf{u}} F(\mathbf{x}_t) \right) - \bar{R}_{p,t}, \quad (28)$$

where

$$\bar{R}_{p,t} := \left(0, 0, \sum_{k=1}^K \bar{r}_{p,t}^k \right), \quad \bar{r}_{p,t}^k := \mathbb{E}[r_{p,t}^k \mid \mathbf{x}_t]. \quad (29)$$

Lemma 2 (Pair-level stochastic deviation bound). *For each collaborative pair p and round t , define*

$$\epsilon_{p,t} := Q_{p,t} - \bar{Q}_{p,t}. \quad (30)$$

Under assumptions 1 2 3, we have

$$\mathbb{E} \|\epsilon_{p,t}\|^2 \leq 2K\sigma_v^2 + 6K\sigma_u^2 + 6KL^2\eta^2(\sigma_u^2 + G_u^2). \quad (31)$$

Lemma 3 (Pair-level residual mean bound). *For each collaborative pair p and round t , let*

$$\bar{R}_{p,t} := \left(0, 0, \sum_{k=1}^K \bar{r}_{p,t}^k \right), \quad \bar{r}_{p,t}^k := \mathbb{E}[r_{p,t}^k \mid \mathbf{x}_t]. \quad (32)$$

Under assumptions 1 2 3, we have

$$\mathbb{E} \|\bar{R}_{p,t}\|^2 \leq 2K^2L^2\eta^2(\sigma_u^2 + G_u^2). \quad (33)$$

Moreover, for any $\alpha_t > 0$,

$$\mathbb{E} \langle \nabla_u F(\mathbf{x}_t), \bar{R}_{p,t}^u \rangle \leq \frac{\alpha_t}{2} \mathbb{E} \|\nabla_u F(\mathbf{x}_t)\|^2 + \frac{K^2L^2\eta^2}{\alpha_t} (\sigma_u^2 + G_u^2). \quad (34)$$

Proof of Theorem 1

Proof. By assumption 1 and the L -smoothness of $F(\cdot)$,

$$F(\mathbf{x}_{t+1}) \leq F(\mathbf{x}_t) + \langle \nabla F(\mathbf{x}_t), \mathbf{x}_{t+1} - \mathbf{x}_t \rangle + \frac{L}{2} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2. \quad (35)$$

Using the pair-level update rule

$$\mathbf{x}_{t+1} - \mathbf{x}_t = -\frac{\eta}{N} \sum_{p=1}^P Q_{p,t}, \quad (36)$$

and taking expectation, we obtain

$$\mathbb{E}[F(\mathbf{x}_{t+1})] \leq \mathbb{E}[F(\mathbf{x}_t)] - \frac{\eta}{N} \sum_{p=1}^P \mathbb{E}\langle \nabla F(\mathbf{x}_t), Q_{p,t} \rangle + \frac{L}{2} \mathbb{E}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2. \quad (37)$$

Write $Q_{p,t} = \bar{Q}_{p,t} + (Q_{p,t} - \bar{Q}_{p,t})$, where $\bar{Q}_{p,t} = \mathbb{E}[Q_{p,t} \mid \mathbf{x}_t]$. Since $\mathbb{E}[Q_{p,t} - \bar{Q}_{p,t} \mid \mathbf{x}_t] = 0$, we have

$$\mathbb{E}\langle \nabla F(\mathbf{x}_t), Q_{p,t} \rangle = \mathbb{E}\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t} \rangle. \quad (38)$$

By Lemma 1,

$$\bar{Q}_{p,t} = \left(K \nabla_v F(\mathbf{x}_t), K \nabla_v F(\mathbf{x}_t), 2K \nabla_u F(\mathbf{x}_t) \right) - \bar{R}_{p,t}. \quad (39)$$

Hence,

$$\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t} \rangle = 2K \|\nabla F(\mathbf{x}_t)\|^2 - \langle \nabla_u F(\mathbf{x}_t), \bar{R}_{p,t}^u \rangle. \quad (40)$$

Applying Lemma 3 with $\alpha_t = K$ yields

$$\mathbb{E}\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t} \rangle \geq \frac{3K}{2} \mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 - KL^2 \eta^2 (\sigma_u^2 + G_u^2). \quad (41)$$

Summing over $p = 1, \dots, P$ and using $N = 2P$, we obtain

$$-\frac{\eta}{N} \sum_{p=1}^P \mathbb{E}\langle \nabla F(\mathbf{x}_t), \bar{Q}_{p,t} \rangle \leq -\frac{3K}{4} \eta \mathbb{E}\|\nabla F(\mathbf{x}_t)\|^2 + \frac{KL^2}{2} \eta^3 (\sigma_u^2 + G_u^2). \quad (42)$$

Next, for the second-order term,

$$\mathbb{E}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 = \frac{\eta^2}{N^2} \mathbb{E} \left\| \sum_{p=1}^P Q_{p,t} \right\|^2. \quad (43)$$

Writing again $Q_{p,t} = \bar{Q}_{p,t} + \epsilon_{p,t}$ with $\epsilon_{p,t} := Q_{p,t} - \bar{Q}_{p,t}$, and noting that $\mathbb{E}[\epsilon_{p,t} \mid \mathbf{x}_t] = 0$, the cross term between the conditional mean part and the stochastic deviation part vanishes. Hence,

$$\mathbb{E}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 = \frac{\eta^2}{N^2} \mathbb{E} \left\| \sum_{p=1}^P \bar{Q}_{p,t} \right\|^2 + \frac{\eta^2}{N^2} \mathbb{E} \left\| \sum_{p=1}^P \epsilon_{p,t} \right\|^2. \quad (44)$$

For the conditional mean term, by Lemma 1, we have

$$\sum_{p=1}^P \bar{Q}_{p,t} = P \left(K \nabla_v F(\mathbf{x}_t), K \nabla_v F(\mathbf{x}_t), 2K \nabla_u F(\mathbf{x}_t) \right) - \sum_{p=1}^P \bar{R}_{p,t}. \quad (45)$$

Hence, by $\|a - b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$,

$$\begin{aligned} \mathbb{E} \left\| \sum_{p=1}^P \bar{Q}_{p,t} \right\|^2 &\leq 2P^2 \left\| (K\nabla_v F(\mathbf{x}_t), K\nabla_v F(\mathbf{x}_t), 2K\nabla_u F(\mathbf{x}_t)) \right\|^2 \\ &\quad + 2\mathbb{E} \left\| \sum_{p=1}^P \bar{R}_{p,t} \right\|^2. \end{aligned} \quad (46)$$

Moreover,

$$\begin{aligned} &\left\| (K\nabla_v F(\mathbf{x}_t), K\nabla_v F(\mathbf{x}_t), 2K\nabla_u F(\mathbf{x}_t)) \right\|^2 \\ &= 2K^2 \|\nabla_v F(\mathbf{x}_t)\|^2 + 4K^2 \|\nabla_u F(\mathbf{x}_t)\|^2 \\ &\leq 4K^2 \|\nabla F(\mathbf{x}_t)\|^2. \end{aligned} \quad (47)$$

Therefore,

$$\mathbb{E} \left\| \sum_{p=1}^P \bar{Q}_{p,t} \right\|^2 \leq 8P^2 K^2 \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 + 2\mathbb{E} \left\| \sum_{p=1}^P \bar{R}_{p,t} \right\|^2. \quad (48)$$

Using

$$\left\| \sum_{p=1}^P \bar{R}_{p,t} \right\|^2 \leq P \sum_{p=1}^P \|\bar{R}_{p,t}\|^2 \quad (49)$$

and Lemma 3, we further obtain

$$\mathbb{E} \left\| \sum_{p=1}^P \bar{Q}_{p,t} \right\|^2 \leq 8P^2 K^2 \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 + 4P^2 K^2 L^2 \eta^2 (\sigma_u^2 + G_u^2). \quad (50)$$

Under pair-wise independent sampling, the cross terms between different collaborative pairs vanish. Applying Lemma 2 and Lemma 3, we obtain

$$\mathbb{E} \|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2 \leq 2K^2 \eta^2 \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 + \frac{\eta^2}{N} (K\sigma_v^2 + 3K\sigma_u^2) + \eta^4 L^2 (\sigma_u^2 + G_u^2) \left(K^2 + \frac{3K}{N} \right). \quad (51)$$

Substituting (42) and (51) into the one-step descent inequality gives

$$\begin{aligned} \mathbb{E}[F(\mathbf{x}_{t+1})] &\leq \mathbb{E}[F(\mathbf{x}_t)] - \left(\frac{3K}{4} \eta - LK^2 \eta^2 \right) \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 \\ &\quad + \frac{LK}{2N} \eta^2 (\sigma_v^2 + 3\sigma_u^2) + \frac{KL^2}{2} \eta^3 (\sigma_u^2 + G_u^2) \\ &\quad + \frac{L^3}{2} \eta^4 (\sigma_u^2 + G_u^2) \left(K^2 + \frac{3K}{N} \right). \end{aligned} \quad (52)$$

When $LK^2 \eta^2 \leq \frac{1}{4} K \eta$, that is $\eta \leq \frac{1}{4LK}$, we have

$$\mathbb{E}[F(\mathbf{x}_{t+1})] \leq \mathbb{E}[F(\mathbf{x}_t)] - \frac{K}{2} \eta \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 + \eta K C_t, \quad (53)$$

where

$$C_t = \frac{L}{2N}\eta(\sigma_v^2 + 3\sigma_u^2) + \frac{L^2}{2}\eta^2(\sigma_u^2 + G_u^2) + \frac{L^3}{2}\eta^3(\sigma_u^2 + G_u^2) \left(K + \frac{3}{N}\right). \quad (54)$$

Summing the above inequality from $t = 0$ to $T - 1$ and using the lower boundedness of $F(\cdot)$ yield

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 \leq \frac{2(F(\mathbf{x}_0) - F(\mathbf{x}_T))}{\eta KT} + 2C_t. \quad (55)$$

Let $\eta = \sqrt{\frac{2(F(\mathbf{x}_0) - F(\mathbf{x}_T))N}{(\sigma_v^2 + 3\sigma_u^2)LKT}}$ into Eq. (54)

we analyze the asymptotic order of each error component in C_t .

The first term, which encapsulates gradient noise and local bias, scales as $\Theta(\eta) = \Theta(T^{-1/2})$, which matches the order of the leading term in Eq. (55).

The second term, induced by second-order effects from local iterations, scales as $\Theta(\eta^2) = \Theta(T^{-1})$, a higher-order term that decays faster than the leading convergence rate.

The third term, which is explicitly introduced by the collaborative alternating update mechanism of SFLMAC, scales as $\Theta(\eta^3) = \Theta(T^{-3/2})$, a negligible higher-order term that vanishes much more rapidly than $O(T^{-1/2})$ as the number of communication rounds T increases.

For sufficiently large T , the overall upper bound in Eq. (55) is asymptotically dominated by the leading-order term $O(T^{-1/2})$, while all higher-order terms—including those introduced by the collaborative update mechanism—become negligible. This leads to the result:

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(\mathbf{x}_t)\|^2 \preceq O\left(\frac{1}{\sqrt{T}}\right),$$

where $\preceq$ denotes the standard order inequality.

Therefore, SFLMAC preserves the standard $O(1/\sqrt{T})$ convergence rate of non-convex stochastic first-order methods.

The collaborative alternating update only contributes higher-order terms that decay faster than the leading rate and thus does not affect the asymptotic convergence behavior of the algorithm, validating the theoretical convergence guarantee of the proposed method.

This completes the proof.

B.3 Proof of Lemma

Proof of Lemma 1

Proof. By the definition of $Q_{p,t}$ and linearity of conditional expectation,

$$\bar{Q}_{p,t} = \left(\sum_{k=1}^K \mathbb{E}[h_{p,t}^{m,k} | \mathbf{x}_t], \sum_{k=1}^K \mathbb{E}[h_{p,t}^{a,k} | \mathbf{x}_t], \sum_{k=1}^K \mathbb{E}[g_{p,t}^{m,k} + g_{p,t}^{a,k} - r_{p,t}^k | \mathbf{x}_t] \right). \quad (56)$$

Under Assumption 3, the stochastic gradients are unbiased estimators of the corresponding global gradients at $\mathbf{x}_t$, i.e.,

$$\mathbb{E}[h_{p,t}^{m,k} \mid \mathbf{x}_t] = \nabla_v F(\mathbf{x}_t), \quad \mathbb{E}[h_{p,t}^{a,k} \mid \mathbf{x}_t] = \nabla_v F(\mathbf{x}_t), \quad (57)$$

and

$$\mathbb{E}[g_{p,t}^{m,k} \mid \mathbf{x}_t] = \nabla_u F(\mathbf{x}_t), \quad \mathbb{E}[g_{p,t}^{a,k} \mid \mathbf{x}_t] = \nabla_u F(\mathbf{x}_t). \quad (58)$$

Substituting the above identities into the definition of $\bar{Q}_{p,t}$ yields (28).

This completes the proof.

Proof of Lemma2

Proof. By the definitions of $Q_{p,t}$ and $\bar{Q}_{p,t}$,

$$\epsilon_{p,t} = \left(\sum_{k=1}^K (h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t)), \sum_{k=1}^K (h_{p,t}^{a,k} - \nabla_v F(\mathbf{x}_t)), \sum_{k=1}^K \zeta_{p,t}^k \right), \quad (59)$$

where

$$\zeta_{p,t}^k := (g_{p,t}^{m,k} - \nabla_u F(\mathbf{x}_t)) + (g_{p,t}^{a,k} - \nabla_u F(\mathbf{x}_t)) - (r_{p,t}^k - \bar{r}_{p,t}^k). \quad (60)$$

Hence,

$$\begin{aligned} \mathbb{E}\|\epsilon_{p,t}\|^2 &= \mathbb{E}\left\|\sum_{k=1}^K (h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t))\right\|^2 + \mathbb{E}\left\|\sum_{k=1}^K (h_{p,t}^{a,k} - \nabla_v F(\mathbf{x}_t))\right\|^2 \\ &\quad + \mathbb{E}\left\|\sum_{k=1}^K \zeta_{p,t}^k\right\|^2. \end{aligned} \quad (61)$$

Since the samples are independently drawn from the same distribution $\mathcal{D}$, the cross terms between different local steps vanish. Therefore,

$$\mathbb{E}\left\|\sum_{k=1}^K (h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t))\right\|^2 = \sum_{k=1}^K \mathbb{E}\|h_{p,t}^{m,k} - \nabla_v F(\mathbf{x}_t)\|^2 \leq K\sigma_v^2, \quad (62)$$

and similarly,

$$\mathbb{E}\left\|\sum_{k=1}^K (h_{p,t}^{a,k} - \nabla_v F(\mathbf{x}_t))\right\|^2 \leq K\sigma_v^2. \quad (63)$$

Since each $\zeta_{p,t}^k$ is formed from the k -th independent sample only and has zero conditional mean given x_t , the cross terms between different local steps vanish. For the upper-layer block, we also have

$$\mathbb{E}\left\|\sum_{k=1}^K \zeta_{p,t}^k\right\|^2 = \sum_{k=1}^K \mathbb{E}\|\zeta_{p,t}^k\|^2. \quad (64)$$

Using $\|a + b - c\|^2 \leq 3\|a\|^2 + 3\|b\|^2 + 3\|c\|^2$, we obtain

$$\mathbb{E}\|\zeta_{p,t}^k\|^2 \leq 3\mathbb{E}\|g_{p,t}^{m,k} - \nabla_u F(\mathbf{x}_t)\|^2 + 3\mathbb{E}\|g_{p,t}^{a,k} - \nabla_u F(\mathbf{x}_t)\|^2 + 3\mathbb{E}\|r_{p,t}^k - \bar{r}_{p,t}^k\|^2. \quad (65)$$

By assumption 3,

$$\mathbb{E}\|g_{p,t}^{m,k} - \nabla_u F(\mathbf{x}_t)\|^2 \leq \sigma_u^2, \quad \mathbb{E}\|g_{p,t}^{a,k} - \nabla_u F(\mathbf{x}_t)\|^2 \leq \sigma_u^2. \quad (66)$$

Moreover, by variance decomposition,

$$\mathbb{E}\|r_{p,t}^k - \bar{r}_{p,t}^k\|^2 \leq \mathbb{E}\|r_{p,t}^k\|^2. \quad (67)$$

By Lemma 3, we have:

$$\mathbb{E}\|r_{p,t}^k\|^2 \leq 2L^2\eta^2(\sigma_u^2 + G_u^2), \quad (68)$$

and consequently,

$$\mathbb{E}\|\zeta_{p,t}^k\|^2 \leq 6\sigma_u^2 + 6L^2\eta^2(\sigma_u^2 + G_u^2). \quad (69)$$

Summing over $k = 1, \dots, K$ yields

$$\mathbb{E}\left\|\sum_{k=1}^K \zeta_{p,t}^k\right\|^2 \leq K\left(6\sigma_u^2 + 6L^2\eta^2(\sigma_u^2 + G_u^2)\right). \quad (70)$$

Combining the above estimates proves (31).

This completes the proof.

Proof of Lemma 3

Proof. By definition,

$$\|\bar{R}_{p,t}\|^2 = \left\|\sum_{k=1}^K \bar{r}_{p,t}^k\right\|^2 \leq K \sum_{k=1}^K \|\bar{r}_{p,t}^k\|^2. \quad (71)$$

Taking expectation on both sides yields

$$\mathbb{E}\|\bar{R}_{p,t}\|^2 \leq K \sum_{k=1}^K \mathbb{E}\|\bar{r}_{p,t}^k\|^2. \quad (72)$$

Since $\bar{r}_{p,t}^k = \mathbb{E}[r_{p,t}^k \mid \mathbf{x}_t]$, Jensen's inequality gives

$$\|\bar{r}_{p,t}^k\|^2 \leq \mathbb{E}[\|r_{p,t}^k\|^2 \mid \mathbf{x}_t]. \quad (73)$$

Hence,

$$\mathbb{E}\|\bar{r}_{p,t}^k\|^2 \leq \mathbb{E}\|r_{p,t}^k\|^2. \quad (74)$$

Therefore,

$$\mathbb{E}\|\bar{R}_{p,t}\|^2 \leq K \sum_{k=1}^K \mathbb{E}\|r_{p,t}^k\|^2. \quad (75)$$

Since the residual term is induced by the upper half-step update, by the Lipschitz continuity of the upper-layer stochastic gradient we have

$$\|r_{p,t}^k\|^2 \leq L^2 \eta^2 \|g_{p,\text{partner}}^k\|^2. \quad (76)$$

Thus,

$$\mathbb{E}\|r_{p,t}^k\|^2 \leq L^2 \eta^2 \mathbb{E}\|g_{p,\text{partner}}^k\|^2. \quad (77)$$

Using

$$g_{p,\text{partner}}^k = (g_{p,\text{partner}}^k - \nabla_u F(\mathbf{x}_t)) + \nabla_u F(\mathbf{x}_t) \quad (78)$$

and $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, we obtain

$$\mathbb{E}\|g_{p,\text{partner}}^k\|^2 \leq 2\mathbb{E}\|g_{p,\text{partner}}^k - \nabla_u F(\mathbf{x}_t)\|^2 + 2\mathbb{E}\|\nabla_u F(\mathbf{x}_t)\|^2. \quad (79)$$

By assumptions 2 3,

$$\mathbb{E}\|g_{p,\text{partner}}^k\|^2 \leq 2\sigma_u^2 + 2G_u^2. \quad (80)$$

Hence,

$$\mathbb{E}\|r_{p,t}^k\|^2 \leq 2L^2 \eta^2 (\sigma_u^2 + G_u^2). \quad (81)$$

Substituting into the previous bound yields

$$\mathbb{E}\|\bar{R}_{p,t}\|^2 \leq K \sum_{k=1}^K 2L^2 \eta^2 (\sigma_u^2 + G_u^2) = 2K^2 L^2 \eta^2 (\sigma_u^2 + G_u^2). \quad (82)$$

This proves (33).

Finally, by Young's inequality,

$$\langle \nabla_u F(\mathbf{x}_t), \bar{R}_{p,t}^u \rangle \leq \frac{\alpha_t}{2} \|\nabla_u F(\mathbf{x}_t)\|^2 + \frac{1}{2\alpha_t} \|\bar{R}_{p,t}^u\|^2. \quad (83)$$

Taking expectation and using (33) gives

$$\mathbb{E}\langle \nabla_u F(\mathbf{x}_t), \bar{R}_{p,t}^u \rangle \leq \frac{\alpha_t}{2} \mathbb{E}\|\nabla_u F(\mathbf{x}_t)\|^2 + \frac{K^2 L^2 \eta^2}{\alpha_t} (\sigma_u^2 + G_u^2). \quad (84)$$

This completes the proof.

C Optimal client pairing selection

Theorem 2. *Let $N = 2P$ clients have computing capacities sorted in ascending order as*

$$x_1 \leq x_2 \leq \dots \leq x_N.$$

Assume that each client initially owns a task of size M , and for each pair (i, j) with $x_i \leq x_j$, the auxiliary client i is allowed to offload an arbitrary portion of its task to the main client j . Then, for a fixed pair (i, j) , the minimum completion time is

$$T_{ij}^* = \frac{2M}{x_i + x_j},$$

achieved by the balanced split $m_i^* = \frac{2Mx_i}{x_i + x_j}$

$$\frac{m_i^*}{x_i} = \frac{2M - m_i^*}{x_j}$$

Consequently, the global pairing problem

$$\min_{\Pi} \max_{(i,j) \in \Pi} T_{ij}^*$$

is equivalent to

$$\max_{\Pi} \min_{(i,j) \in \Pi} (x_i + x_j).$$

Among all pairing schemes, the strongest-weakest pairing

$$(1, N), (2, N-1), \dots, (P, P+1)$$

is optimal.

Proof. For any pair (i, j) with $x_i \leq x_j$, let $m_i \in [0, M]$ denote the workload processed by client i . Then client j processes workload $2M - m_i$, and the pair completion time is

$$T_{ij}(m_i) = \max \left\{ \frac{m_i}{x_i}, \frac{2M - m_i}{x_j} \right\}.$$

Since $\frac{m_i}{x_i}$ is increasing in m_i and $\frac{2M - m_i}{x_j}$ is decreasing in m_i , the minimum is attained when the two terms are equal:

$$\frac{m_i}{x_i} = \frac{2M - m_i}{x_j}.$$

Solving gives

$$m_i^* = \frac{2Mx_i}{x_i + x_j}$$

and hence

$$T_{ij}^* = \frac{2M}{x_i + x_j}.$$

Therefore, for any pairing policy Π ,

$$T(\Pi) = \max_{(i,j) \in \Pi} T_{ij}^* = \max_{(i,j) \in \Pi} \frac{2M}{x_i + x_j} = \frac{2M}{\min_{(i,j) \in \Pi} (x_i + x_j)}.$$

Thus, minimizing $T(\Pi)$ is equivalent to maximizing

$$\min_{(i,j) \in \Pi} (x_i + x_j).$$

We now prove that the strongest-weakest pairing

$$\Pi^* = \{(1, N), (2, N-1), \dots, (P, P+1)\}$$

maximizes the minimum pair sum.

Let Π be any pairing. Since client 1 must be paired with some client k , we have

$$x_1 + x_k \leq x_1 + x_N.$$

Hence,

$$\min_{(i,j) \in \Pi} (x_i + x_j) \leq x_1 + x_k \leq x_1 + x_N.$$

So no pairing can have minimum pair sum larger than $x_1 + x_N$.

Now we show that there exists an optimal pairing containing the pair $(1, N)$. Suppose in some optimal pairing, client 1 is paired with a and client N is paired with b , where $a \neq N$ and $b \neq 1$. Then these two pairs are $(1, a)$ and (b, N) , with $x_1 \leq x_b \leq x_a \leq x_N$ after relabeling if necessary. Replace them by $(1, N)$ and (a, b) . The new pair sums are

$$x_1 + x_N, \quad x_a + x_b.$$

Compared with the original pair sums

$$x_1 + x_a, \quad x_b + x_N,$$

we have

$$x_1 + x_N \geq x_1 + x_a, \quad x_a + x_b \geq x_b + x_N.$$

Therefore, the minimum of the two new pair sums is no smaller than the minimum of the two original pair sums. Hence this replacement does not decrease the minimum pair sum. So there always exists an optimal pairing containing $(1, N)$.

After fixing the pair $(1, N)$, the remaining clients are

$$x_2 \leq x_3 \leq \dots \leq x_{N-1},$$

which form the same problem of size $N - 2$. Repeating the same argument recursively shows that there exists an optimal pairing containing

$$(1, N), (2, N - 1), \dots, (P, P + 1).$$

Therefore, the strongest-weakest pairing maximizes the minimum pair sum, and equivalently minimizes the per-round completion time.

This completes the proof.